

Session 8

Q1. Write a PL/SQL block that declares two variables: one for a user's name (VARCHAR2) and one for their age (NUMBER). Assign values and print both using DBMS_OUTPUT.PUT_LINE.

The screenshot shows the Oracle FreeSQL web interface. The main editor contains the following PL/SQL code:

```
1 DECLARE
2   user_name VARCHAR2(50);
3   user_age NUMBER;
4 BEGIN
5   user_name := 'Purvang';
6   user_age := 21;
7   DBMS_OUTPUT.PUT_LINE('Name: ' || user_name);
8   DBMS_OUTPUT.PUT_LINE('Age: ' || user_age);
9 END;
```

The 'Script output' tab shows the results of the execution:

```
user_age NUMBER;
BEGIN...
Show more...

Name: Purvang
Age: 21

PL/SQL procedure successfully completed.
Elapsed: 00:00:00.006
```

Q2. Create a PL/SQL block that uses an IF statement to check if a given order amount is above 500. If yes, print 'Eligible for free delivery', else print 'Delivery charges apply'.

The screenshot shows the Oracle FreeSQL web interface. The main editor contains the following PL/SQL code:

```
1 DECLARE
2   user_order NUMBER := 850;
3 BEGIN
4   IF user_order > 500 THEN
5     DBMS_OUTPUT.PUT_LINE('Eligible for free delivery');
6   ELSE
7     DBMS_OUTPUT.PUT_LINE('Delivery charges apply');
8   END IF;
9 END;
```

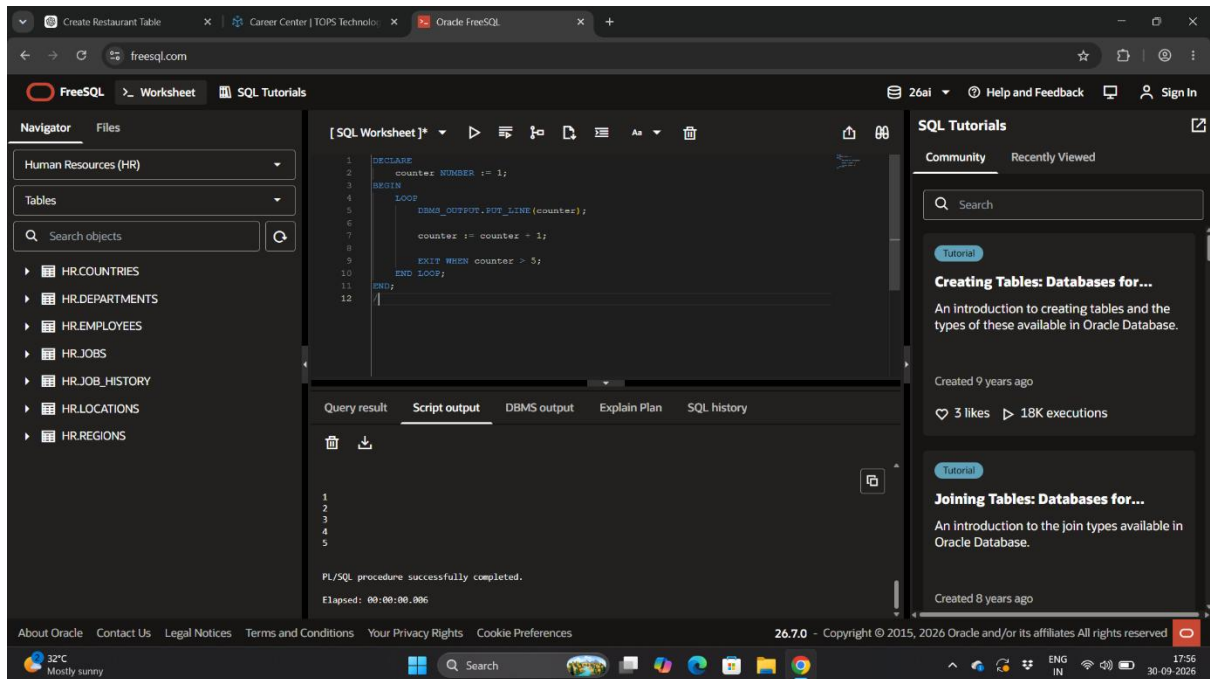
The 'Script output' tab shows the results of the execution:

```
user_order NUMBER := 850;
BEGIN
  IF user_order > 500 THEN...
Show more...

Eligible for free delivery

PL/SQL procedure successfully completed.
Elapsed: 00:00:00.006
```

Q3. Write a PL/SQL block that uses a simple LOOP to print the numbers 1 to 5 using DBMS_OUTPUT.PUT_LINE. Hint: Use a counter variable and EXIT WHEN condition inside the loop.



The screenshot shows the Oracle FreeSQL web interface. The main editor displays the following PL/SQL code:

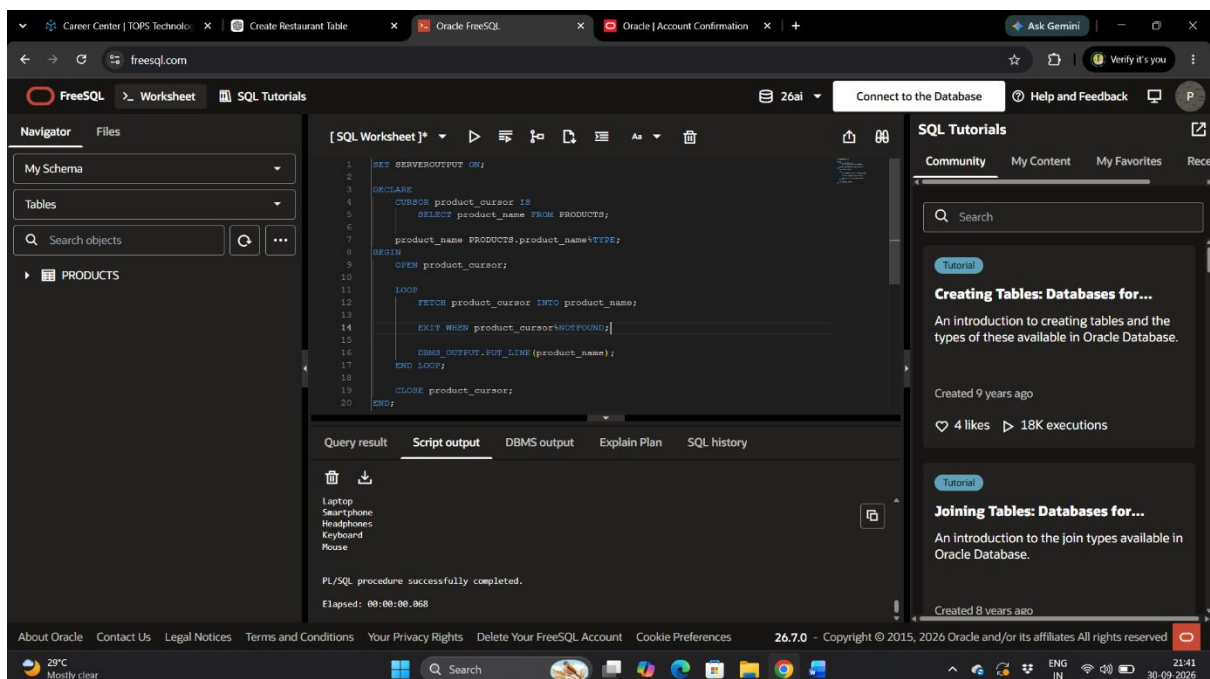
```

1 DECLARE
2   counter NUMBER := 1;
3 BEGIN
4   LOOP
5     DBMS_OUTPUT.PUT_LINE(counter);
6     counter := counter + 1;
7     EXIT WHEN counter > 5;
8   END LOOP;
9 END;

```

The interface also shows a sidebar with a Navigator pane containing a tree view of database objects (Human Resources (HR), Tables, etc.) and a SQL Tutorials pane on the right with links to 'Creating Tables: Databases for...' and 'Joining Tables: Databases for...'. The bottom status bar indicates the PL/SQL procedure was successfully completed.

Q4. Declare a cursor in PL/SQL to select all product names from a table called PRODUCTS. Fetch each product name and print it using DBMS_OUTPUT.PUT_LINE. Hint: Assume the PRODUCTS table has a column PRODUCT_NAME.



The screenshot shows the Oracle FreeSQL web interface. The main editor displays the following PL/SQL code:

```

1 SET SERVEROUTPUT ON;
2 DECLARE
3   CURSOR product_cursor IS
4     SELECT product_name FROM PRODUCTS;
5   product_name PRODUCTS.product_name%TYPE;
6 BEGIN
7   OPEN product_cursor;
8   LOOP
9     FETCH product_cursor INTO product_name;
10    EXIT WHEN product_cursor%NOTFOUND;
11    DBMS_OUTPUT.PUT_LINE(product_name);
12  END LOOP;
13 CLOSE product_cursor;
14 END;

```

The interface also shows a sidebar with a Navigator pane containing a tree view of database objects (My Schema, Tables, etc.) and a SQL Tutorials pane on the right with links to 'Creating Tables: Databases for...' and 'Joining Tables: Databases for...'. The bottom status bar indicates the PL/SQL procedure was successfully completed.

Q5. Write a PL/SQL block that tries to divide two numbers entered as variables. If a division by zero occurs, catch the exception and print 'Cannot divide by zero'. Otherwise, print the result.

The screenshot displays the Oracle FreeSQL web application. The main area is a SQL Worksheet where a PL/SQL block is defined and executed. The block declares two variables, `num1` and `num2`, with values 100 and 0 respectively. It then attempts to calculate `num1 / num2`. Since `num2` is zero, an exception is raised, and the message 'Cannot divide by zero' is printed. The execution status is 'PL/SQL procedure successfully completed.' and the elapsed time is 00:00:00.006.

```
1 DECLARE
2   num1 NUMBER := 100;
3   num2 NUMBER := 0;
4   result NUMBER;
5 BEGIN
6   result := num1 / num2;
7   DBMS_OUTPUT.PUT_LINE('Result: ' || result);
8
9 EXCEPTION
10  WHEN ZERO_DIVIDE THEN
11    DBMS_OUTPUT.PUT_LINE('Cannot divide by zero');
12 END;
```

The output pane shows the following results:

```
SQL Worksheet
Script output
DBMS output
Explain Plan
SQL history

Cannot divide by zero

PL/SQL procedure successfully completed.
Elapsed: 00:00:00.006
```

The right-hand sidebar contains SQL Tutorials, including 'Creating Tables: Databases for...' and 'Joining Tables: Databases for...'. The bottom of the page shows the Oracle logo, navigation links, and the version number 26.7.0.